

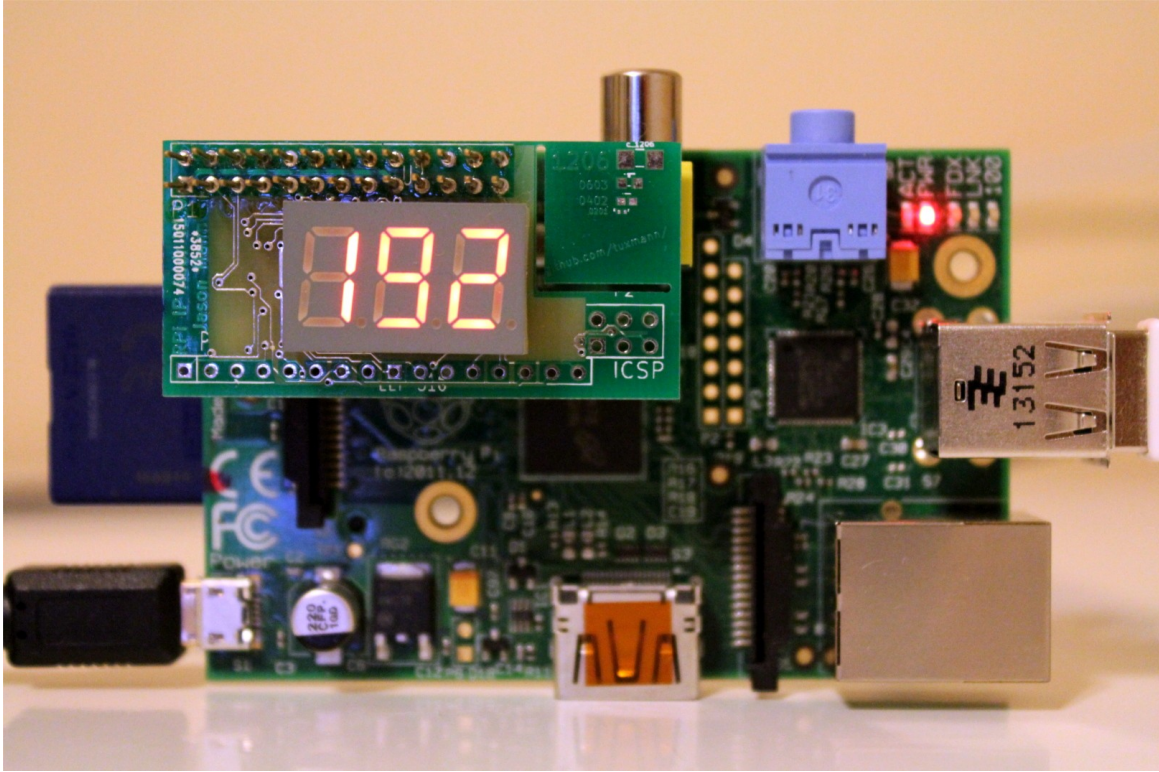
Pi IP Address add-on board for the Raspberry Pi

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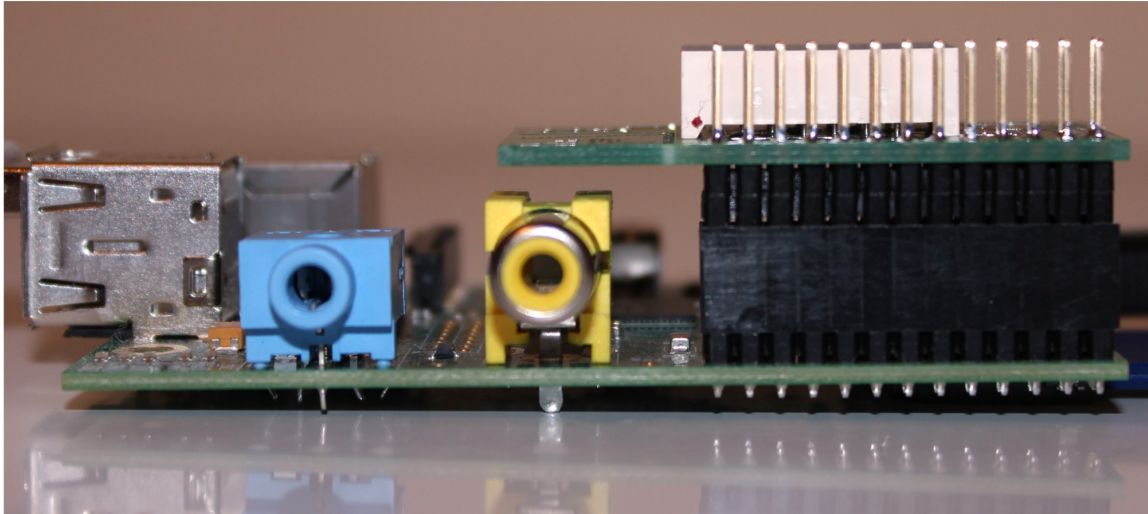
Software License: GPL V3



The Pi IP Address board makes it easier to know the IP Address of your Raspberry Pi when it's connected to a network. This board is compatible with all Raspberry Pi computers as well as other single board computers that are pin compatible. The board sits 0.6 inches (or 1.5cm) above the Raspberry Pi. The tops of the header pins go up to 1 inch (or 2.5cm). The add-on board uses an ATtiny88 by Atmel to drive a 3 digit 7 segment LED display. There are 16 pins from the ATtiny88 that are brought out to the edge of the board to give the user the opportunity to increase the functionality of the add on board.

Setting up the Raspberry Pi

Ensure that the Raspberry Pi is powered off before installing the add-on board. Install the add on board so that all of the 26 original pins are connected as seen in the picture below. Take care to not offset the pins or it may cause serious damage to the Pi IP Address add-on board or your Raspberry Pi. Plug in the Raspberry Pi and allow it to boot up. You will notice that the Pi IP Address board is counting from 001 to 004. This is acceptable because it hasn't yet received the IP address from the Raspberry Pi.



Log in to the Raspberry Pi. I will be logging in under the username of pi. Replace 'pi' with your username if needed.

Install the required software to communicate with the IP Aand clone the repository

```
sudo apt-get update && sudo apt-get install git python-smbus i2c-tools  
git clone https://github.com/tuxmann/Raspberry_IP_Addr.git
```

Install communications software to talk to the Pi IP Address add-on board by editing the modules file.

```
sudo nano /etc/modules
```

Add these lines to the modules file:

```
i2c-bcm2708  
i2c-dev
```

Ctrl + X and save the file

Enable the I2C communcation

```
sudo nano /etc/modprobe.d/raspi-blacklist.conf
```

Comment out this line by adding a '#' character

```
#blacklist i2c-bcm2708
```

Ctrl + X and save the file

Gain root access and add the python script to run at startup

```
sudo su  
crontab -e
```

Add this line to the end of the file (change pi to username you're logged in as):

```
@reboot python /home/pi/Raspberry_IP_Addr/hostip.py
```

Reboot the Raspberry Pi

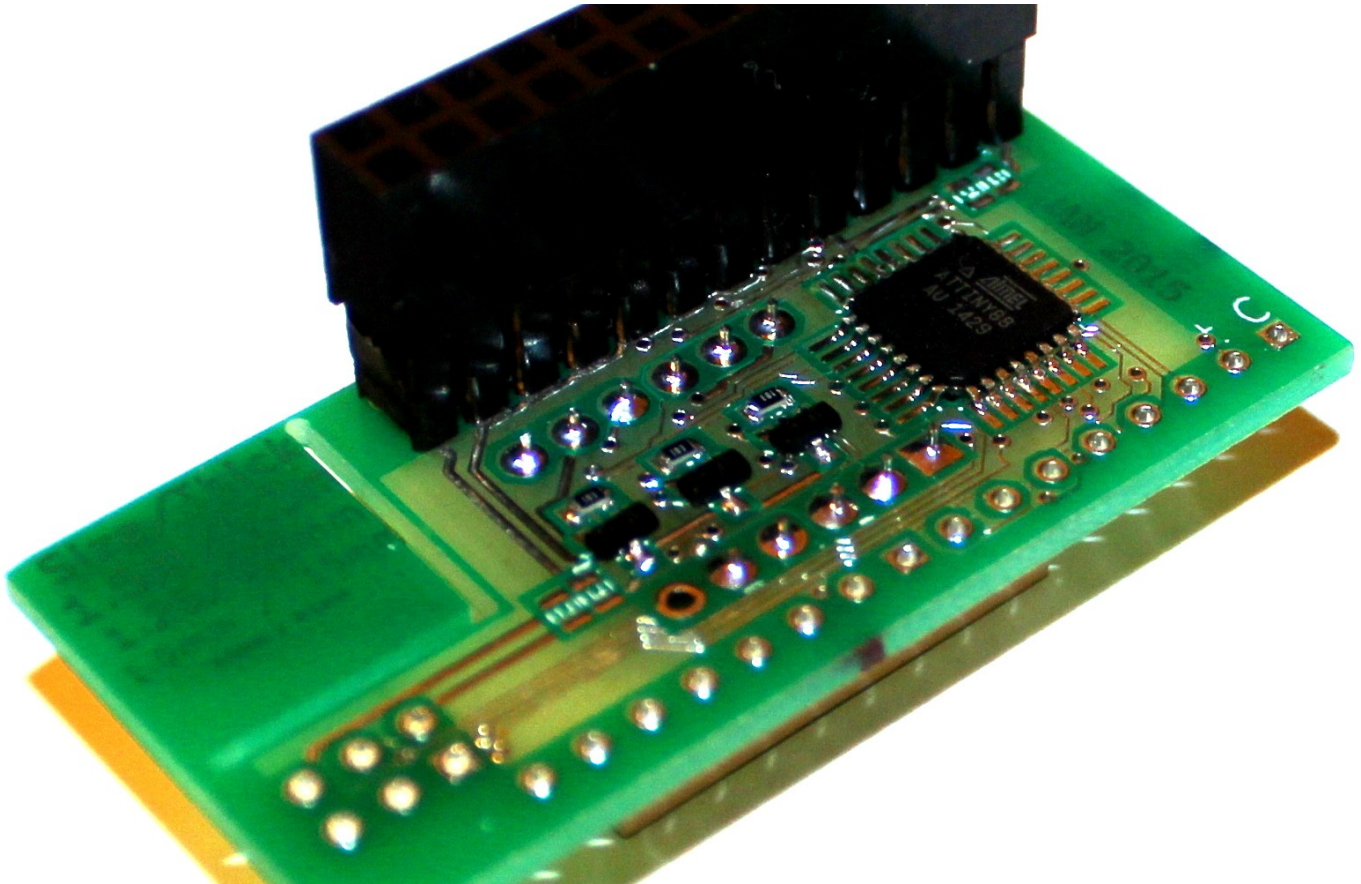
```
sudo reboot
```

Once the Raspberry Pi finishes rebooting and if it's plugged into a network connection (wireless or not) it will display the IP address. The display will show “255.000.255.000” if it's not connected to a network. If wireless and wired are both connected, it will not display either.

To verify that the program is working correctly, you can run the command below. Navigate to the directory that holds the `hostip.py` file.

```
cd /path/to/hostip.py  
sudo python hostip.py
```

Programming the add-on board



If you want to reprogram the Pi IP Address add-on board to do more (or less) than what I programmed it for, these are the instructions that will get you up & running. You will also need four wires connected to the 6 pin ICSP port in order to program the Pi IP Address.



Setting up the Raspberry Pi as an AVR Programmer.

This assumes that you don't already have the *avrdude* program already installed. If you already have a version installed, you may experience problems programming the ATtiny88.

Install necessary software to compile the code from Kevin Cuzner's repository.

```
sudo apt-get install bison autoconf flex
```

Remove the SPI interface from the raspi-blacklist.conf by commenting it out

```
sudo nano /etc/modprobe.d/raspi-blacklist.conf
```

```
#blacklist spi-bcm2708
```

Ctrl + X and save the file

Clone the repository

```
git clone https://github.com/kcuzner/avrdude.git
```

Run the “./bootstrap” script

```
cd avrdude/avrdude/
```

```
./bootstrap
```

```
./configure (Ensure that “ENABLED linuxspi” shows up at the bottom)
```

```
make
```

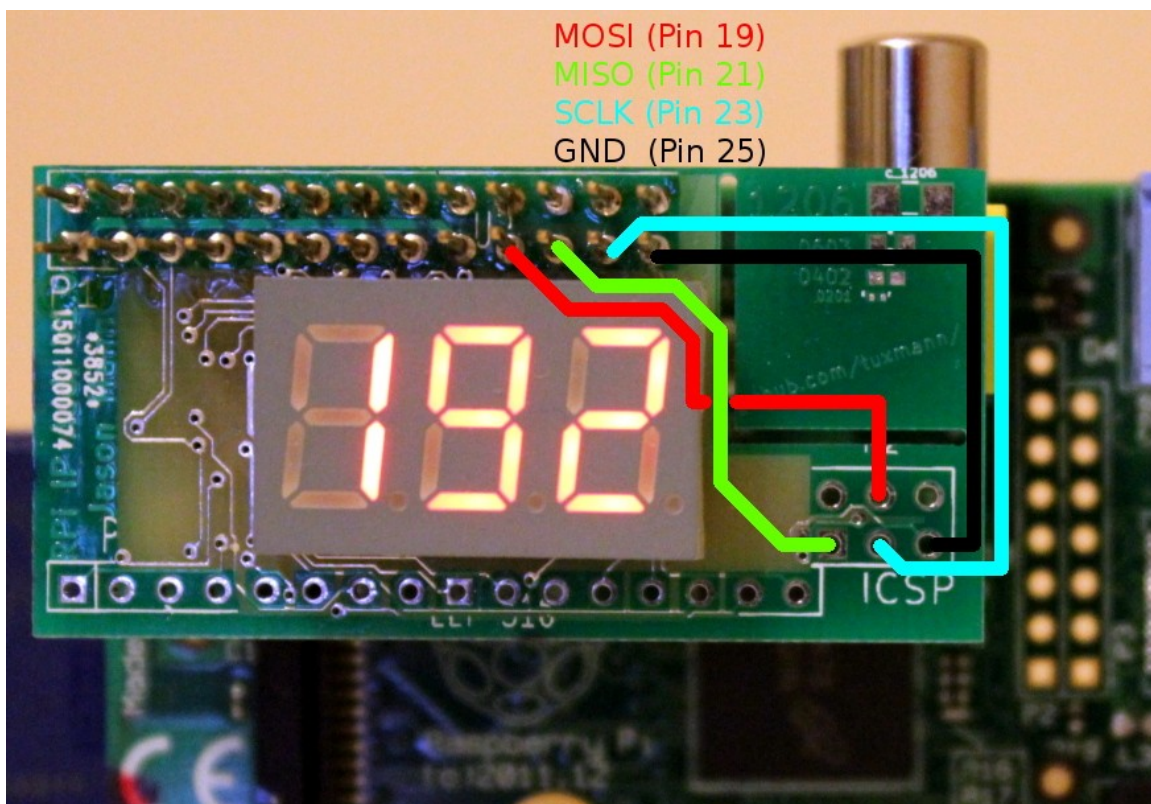
```
sudo make install
```

Verifying that linuxspi is a valid programmer

```
avrdude -c ?type
```

Verify that you can reprogram the ATtiny88 by connecting the wires and running the command below. Use wires that have a male connection on one side and female on the opposite side. Be careful to connect the wires to their intended pins and vias in order to prevent damage. The GND ground wire will connect to the RESET pin of the ATtiny88 and it will be ready to program. Once you're finished programming, disconnect the ground wire to let the ATtiny88 out of reset. The sample program will count down from four instead of counting up to four.

```
sudo avrdude -c linuxspi -p t88 -P /dev/spidev0.0 -U flash:w:pi_wohs.hex
```



Programming commands

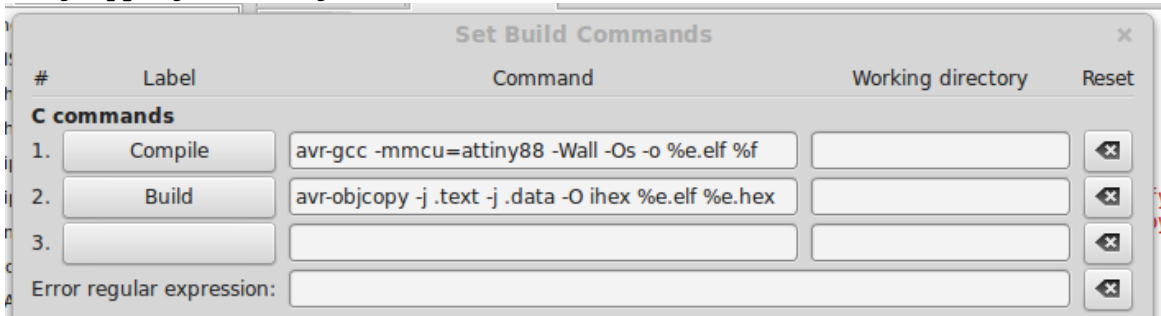
For my programming setup, I used Geany & then I pushed the final .hex file to the Raspberry Pi with Filezilla. Once the file is on the Raspberry Pi, ground the reset line on the ICSP port and then run the command below.

```
sudo avrdude -c linuxspi -p t88 -P /dev/spidev0.0 -U flash:w:your_file.hex
```

Geany build setup:

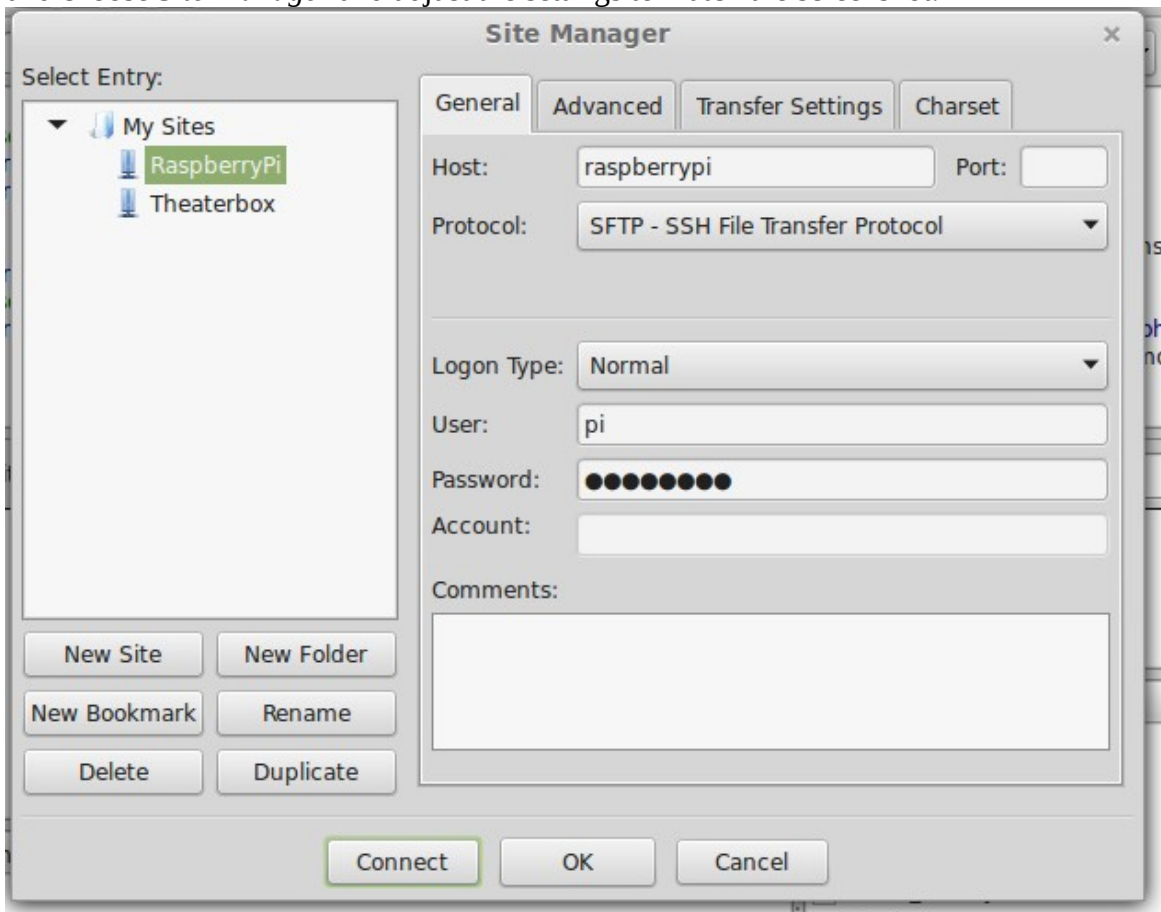
Choose *Build* and then *Set Build Commands* and put in lines below as seen in the screenshot.

```
avr-gcc -mmcu=attiny88 -Wall -Os -o %e.elf %f
avr-objcopy -j .text -j .data -O ihex %e.elf %e.hex
```



Filezilla SFTP setup:

Go to *File* and choose *Site Manager* and adjust the settings to match the screenshot.



Resources

<http://www.atmel.com/devices/attiny88.aspx>

<http://kevincuzner.com/2013/05/27/raspberry-pi-as-an-avr-programmer/>

<http://pengu.student.utwente.nl/wordpress/?p=136>

To Do:

Turn the install instructions into a bash script

Add the idle animations

Comment & Clean up code

Documentation

Create a BOM

~~Ensure that Pi IP Address cycle times are slow enough.~~